TUTORIAL 2 (EXERCISE 2)

Question 1

a) A researcher collected data on the number of hours students spent studying for an exam over a week as shown in Table 1 below.

Table 1.

Study Hours (hours)	Frequency
0-5	8
6-10	12
11-15	15
16-20	10
21-25	5

Calculate

i. Mean [2 marks]

ii. Median [4 marks]

iii. Mode [3 marks]

Question 2

Table 2 gives the number of books read by 100 students surveyed in 2022.

Table 2: Number of books read by 100 students

Number of books	Number of students
0−5	16
6−10	28
11−15	20
16−20	17
21−25	19

From Table 2, calculate;

a) Mean (2 marks)

b) Median (4 marks)

c) Mode (4 marks)

QUESTION 3 **(10 MARKS)**

a) Explain a positively/right skewed distribution (draw a diagram) and show the mean, median, and mode distributions in the diagram.

(4 marks)

b) Consider the data in Table 3, where the lengths of 30 fish caught in a competition were measured to the closest centimetre (cm).

Table 3: Lengths of 30 Fish

25	22	16	17	30	32	37	40	31	23
38	24	27	33	21	28	28	26	34	28
32	26	33	29	27	26	35	28	30	35

i) Calculate the mean length of the fish.

(2 marks)

ii) Calculate the variance and standard deviation for the given data.

(4 marks)